

ORIGINAL ARTICLE

Model-informed precision dosing of carboplatin in cancer patients by leveraging myelosuppression data from electronic health records

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Abstract

Aims: This study aimed to develop and validate a population pharmacokinetic-pharmacodynamic (pop-PK-PD) model to describe carboplatin-induced myelosuppression in cancer patients and support dose individualization.

Methods: Data from 580 cancer patients treated with carboplatin at Amsterdam UMC between 2019 and 2022 were used for model development, focusing on lung, gynaecological and gastric/oesophageal cancers. Platelet (PLT) and neutrophil (NT) counts, along with patient-specific covariates (e.g., age, serum albumin, eGFR), were extracted from Electronic Health Records and used in the analysis. Given the absence of pharmacokinetic (PK) samples, PK parameters were derived from a literature carboplatin pop-PK model. Model applicability to inform personalized carboplatin dosing was evaluated on a separate cohort of 210 patients treated between 2022 and 2024 in the same centre.

Results: Two joint Friberg models effectively described carboplatin-induced myelosuppression. Serum albumin, eGFR and paclitaxel and pemetrexed co-medications were included in the final model. On the test cohort, >85% of NT and >87% of PLT observations fell within the 90% confidence interval of Bayesian model predictions, confirming that the model can support dose adjustments for subsequent treatment cycles. An example of model-based dose adjustments is also presented with a simulation study.

Conclusions: The pop-PK-PD model demonstrated strong performance in describing and predicting carboplatin-induced myelosuppression, thus providing a valuable strategy for dose personalization. Further refinements and validation steps are needed before integrating such an approach into clinical workflows.

KEYWORDS

Bayesian inference, carboplatin-dosing, carboplatin-induced myelosuppression, model-informed precision dosing, PK-PD modeling, real-world data, toxicity-based dosing

Paolo Magni and Imke Bartelink shared authorship.

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1 | INTRODUCTION

Carboplatin is a widely used cytotoxic agent in the treatment of several types of cancers, including lung, gynaecological and gastric/oesophageal cancers, either in monotherapy or in combination with other chemotherapeutic and/or immunotherapeutic agents.^{1–4} Despite its established role in oncology, myelosuppression, i.e., excessive reduction of blood cells levels, remains the main dose-limiting toxicity for carboplatin, with thrombocytopenia and neutropenia frequently leading to dose reductions, treatment delays or interruptions.^{5,6}

Currently, carboplatin dosing is guided by the Calvert equation^{7,8} which aims to achieve a target exposure based on patient renal function. The equation originally used creatinine clearance (CrCl), estimated via the Cockcroft and Gault (CG) equation,⁹ as a surrogate for the glomerular filtration rate (GFR). Although more accurate methods for estimating GFR, such as the CKD-EPI equation,¹⁰ are available, CrCl estimate using the CG equation remains the most commonly used approach in clinical practice and in the application of the Calvert formula. While dose adjustments based on the Calvert equation represent the gold-standard for carboplatin treatment, this empirical approach does not fully account for inter-individual variability (IIV), and may result in underexposure or excessive toxicity, particularly in patients with altered renal function or poor bone marrow reserve.

Carboplatin dose individualization strategies based on periodic monitoring of drug exposure (therapeutic drug monitoring, TDM) and/or efficacy and safety biomarkers can maximize treatment efficacy and reduce the onset of myelosuppression, thereby addressing the limitation of empirical dosing protocols. Although TDM-based approaches have been investigated in specific clinical contexts and selected patient populations,^{11–14} such as paediatric cohorts and high-dose chemotherapy regimens, they have not yet been widely implemented in routine clinical practice. Based on current guidelines,¹⁵ it is reasonable to adjust carboplatin dose to reach the maximum tolerated dose for each patient, enabling dose intensification when feasible and avoiding severe toxicity events (i.e., CTCAE grade ≥ 3); mild-to-moderate toxicity events (CTCAE grade ≤ 2) are, indeed, considered clinically acceptable.¹⁶

Population pharmacokinetics-pharmacodynamics (pop-PK-PD) modelling and simulation play a central role in supporting personalized treatments, thus creating a novel approach called Model Informed Precision Dosing (MIPD).^{17,18} MIPD uses a pop-PK-PD model as Bayesian priors and combines them with patient data (i.e., biomarkers samples collected during treatment and individual covariates) to estimate the posterior distribution of patient-specific PK-PD parameters, thereby enabling the prediction and the optimization of patient future treatment.^{17–19} As patient characteristics and treatment contexts in routine clinical practice may substantially differ from those observed in controlled clinical trial settings,^{20–22} it is essential that pop-PK-PD models intended for MIPD applications are developed on data that reflect these real-world conditions.^{17–19} In this context, Electronic health records (EHRs) data represent a useful source of information to develop robust pop-PK-PD models for the targeted patients.^{23–25}

What is already known about this subject

- Chemotherapy often leads to myelosuppression, which can harm patient health and lead to treatment interruptions and poor response.
- Carboplatin dosing via the Calvert equation can cause severe toxicity or reduced efficacy due to patient variability.
- Pop-PK-PD models describe treatment response variability and enable Model-Informed Precision Dosing to optimize efficacy/toxicity trade-off.

What this study adds

- A pop-PK-PD model for carboplatin-induced myelosuppression was developed and validated using real-world data of 791 cancer patients.
- Key covariates describing inter-individual variability were serum albumin, renal function biomarkers and concomitant chemotherapeutics.
- This study remarks on the potential of model-based dose personalization for carboplatin and defines future developments for its clinical application.

The aim of this work is to develop and validate a pop-PK-PD model of carboplatin-induced myelosuppression to support dose individualization in cancer patients. Furthermore, we aimed to use model simulations to demonstrate how MIPD of carboplatin can be applied in the clinical setting. In particular, the target population was focused on the most frequent tumour diagnoses in which carboplatin is a standard treatment (i.e., lung, gynaecological and oesophageal/gastric neoplasm). To this end, real-world data from Amsterdam UMC EHRs were leveraged to extract and define model training and test sets. Obtained results showed that the implemented model performs well in both describing and predicting carboplatin-induced myelosuppression, and, consequently, it can be considered as a valid tool to apply a MIPD of carboplatin in this centre.

2 | METHODS

2.1 | Dataset building

Data from cancer patients receiving carboplatin treatment were retrospectively and anonymously retrieved from the Epic EHR database of the Amsterdam UMC. The Medical Ethics Review Committee at Amsterdam UMC, location VUmc, declared that the study was not subject to the Medical Research Involving Human Subjects Act (Non-WMO declaration 20 220 174). Demographics information and other

TABLE 1 Statistical summary of the training population statistics (N = 580).

Sample number and continuous covariates	
Observations	Median [95% C.I.]
Platelets (PLT)	6 [2, 14.5]
Neutrophils (NT)	5 [1, 12]
Covariate [units]	Median [95% C.I.]
Age [years]	68 [39, 84]
Weight* [Kg]	73 [50, 116]
Height [m]	1.72 [1.55, 1.92]
BSA* [m ²]	1.86 [1.49, 2.38]
BMI* [Kg/m ²]	24.54 [17.33, 38.40]
CKD-EPI** eGFR* [ml/min]	85.89 [37.80, 115.91]
CG eGFR* [ml/min]	87.57 [38.10, 115]
Serum Creatinine* [μmol/L]	72 [45, 139]
Serum Albumin* [g/L]	37 [25, 47]
LDH* [U/L]	206 [135, 599]
Categorical covariates	
Covariate	Values (Number - frequency, %)
Gender	- Female (284–49%) - Male (296–51%)
Cancer Diagnosis	- Lung (301–52%) - Gynaecology (93–16%) - Oesophageal/gastric (186–32%)
Curative Aim annotated on EHR	- Curative (354–61%) - Palliative (226–39%)
Concomitant Radio Therapy	- Yes (180–31%) - No (400–69%)
Concomitant Capecitabine Medication	- Yes (12–2%) - No (568–98%)
Concomitant Paclitaxel Medication	- Yes (296–51%) - No (284–49%)
Concomitant Pemetrexed Medication	- Yes (406–70%) - No (174–30%)
Concomitant Etoposide Medication	- Yes (64–11%) - No (516–89%)
Number of concomitant Immunotherapies	0 (412–71%) 1 (156–27%) 2 (12–2%)

*time-varying covariates.

**computed using the CKD-EPI 2009 equation.¹⁰

patient characteristics are shown in Table 1. Longitudinal information on carboplatin dosing and laboratory results, including platelets (PLT) and neutrophils (NT) counts, was collected. Figure 1 represents the pipeline implemented to develop the final dataset. A first extraction from the Epic EHR database was performed considering the cancer patients treated with carboplatin in the January 2019–June 2022 timeframe; these data were used to build up the training set for the pop-PK-PD model describing carboplatin-induced myelosuppression. Then, a second extraction considering patients treated in the June 2022–

January 2024 timeframe was performed on the same database to construct the cohort used to evaluate the model applicability in a MIPD framework (i.e., test set). Raw data were filtered by cascading the following criteria. Only patients with the most frequent tumour diagnoses in which carboplatin is a standard treatment were considered, i.e. lung, gynaecological and oesophageal/gastric neoplasms. Patients included in the training set were removed from the test set. Patients with completely missing information (i.e., absence of any measurement for at least one of the longitudinal laboratory results) were excluded (11%) to avoid imputing clinically essential information. This exclusion ensured data quality and falls within accepted thresholds (~10%). Then, the time-profiles of patients PLT, NT and dosage were graphically checked to detect the presence of inconsistencies. Outliers, biologically implausible/abnormal values of PLT, NT and covariates, were removed. Furthermore, multiple measurements of PLT and NT associated to the same timepoint were removed if their range (i.e., difference between maximum and minimum values) was >25 Gi/L for PLT and >0.5 Gi/L for NT, otherwise the mean point was considered.

In conclusion, the final training dataset included 580 patients, with a total of 3635 PLT observations and 2726 NT observations. The test dataset included 210 patients, for a total of 699 PLT and 783 NT observations. Tables 1 and S1.1 summarize the sample numbers and covariates distributions in the training and test populations, respectively.

2.2 | Carboplatin-induced myelosuppression pop-PK-PD model

The basic structure of the implemented carboplatin-induced myelosuppression pop-PK-PD model leverages both carboplatin pop-PK model implemented by Ekhardt et al. (2006)²⁶ and the semi-mechanistic PD model of chemotherapy-induced myelosuppression developed by Friberg et al.²⁷ (Figure S1.2). Briefly, Ekhardt et al.²⁶ described carboplatin PK following intravenous infusion in a Dutch population by using a two-compartment model, with carboplatin clearance (CL) being affected by patient age, serum creatinine concentration and weight (further details in Supplementary Materials S1). Given the sparsity of data and the absence of PK samples collected in the clinical routine setting, individual PK parameters were fixed to the population values found in the original work²⁶ and adjusted by patient covariates.

Carboplatin myelosuppression on PLT and NT was described with two Friberg models²⁷ (details in Supplementary Materials S1). These models describe the maturation of PLT and NT progenitor cells in the bone marrow (i.e., Proliferating compartment) through N (usually $N=3$) transit compartments characterized by a transition rate, kTr , which is defined as a function of the mean maturation time, MTT , $kTr = (N+1)/MTT$. A further compartment is used to represent circulating mature PLT and NT, which are cleared from the blood at a rate $kCirc$. The proliferation rate of PLT and NT precursor is characterized by the parameter $kProl$, which is regulated via a feedback mechanism (i.e., $(\frac{Circ}{Circ_0})^\gamma$) emulating homeostatic mechanisms present in the body to maintain PLT and NT to a baseline value ($Circ_0$). At steady state,

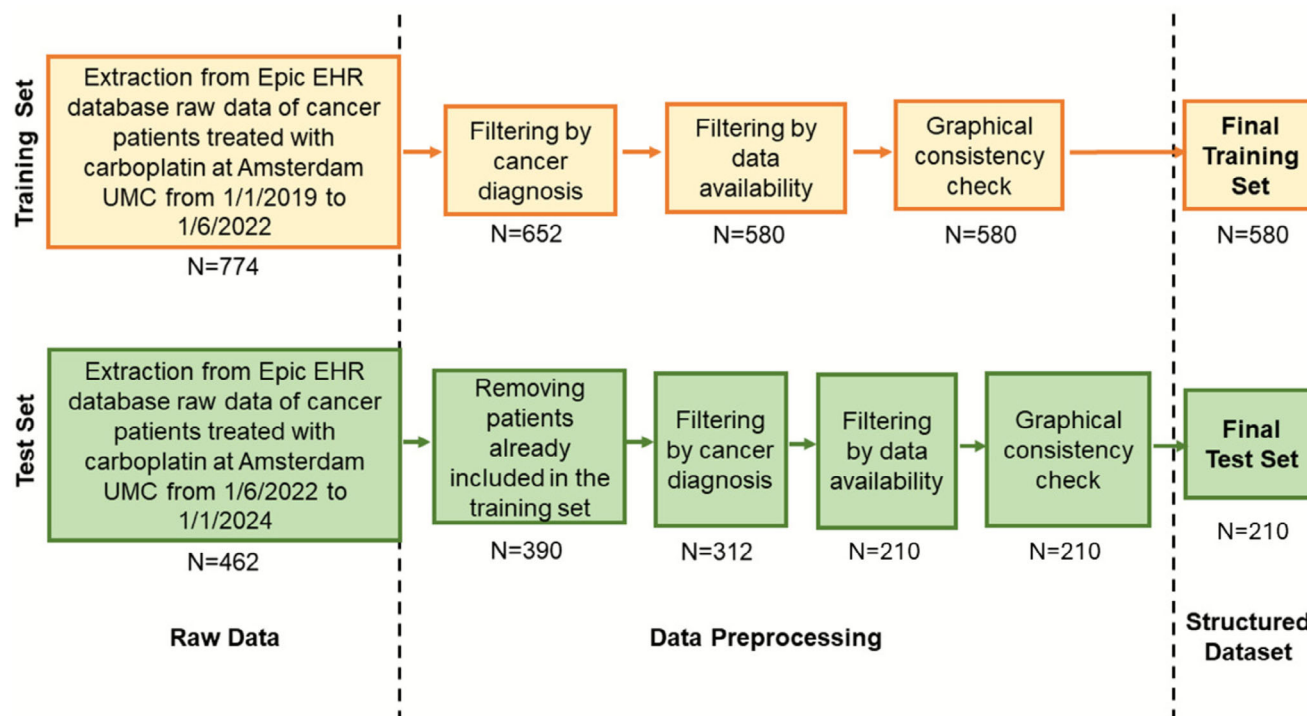


FIGURE 1 Flowchart of the preprocessing steps implemented to build up the training and test sets from raw data extracted from epic HER database. Applied filtering criteria are described in the dataset building subsection of the methods.

$kProl = kCirc = kTr$. Chemotherapy toxicity is modelled with an inhibition of the proliferation mechanism driven by drug concentration. In this case, a linear relationship (i.e., $Slope \cdot Conc$) was used to describe carboplatin reduction of the proliferation rate.

2.3 | Model development and covariate analyses

The training set of 580 patients was used to develop the final pop-PK-PD model starting from the basic structural version of the Friberg myelosuppression models (Figure S1.2). IIV was considered on PLT and NT parameters with a log-normal model; a covariate search was manually performed to explain it.

The covariates depicted in Table 1 were considered based on their known or hypothesized relationships with carboplatin treatment outcomes and clinical availability. Demographics information (i.e., gender, age, weight), and GFR (estimated with both CKD-EPI 2009¹⁰ and CG equations) were included as associated with carboplatin PK.^{9,28,29} Concomitant chemotherapies were considered as related to treatment outcomes and/or myelotoxicity.^{30–36} In addition, serum albumin^{37,38} and lactate dehydrogenase (LDH) levels,^{39–42} as indicators of tumour burden and disease aggressiveness, together with concomitant immunotherapies, considered for their potential clinical benefit, were also tested as possible covariates, although they showed only limited direct association with hematologic toxicity.^{43,44} Pearson correlation and ANOVA tests^{45,46} were used to test the inclusion of continuous and categorical covariates, respectively (significance level was fixed to 1×10^{-3}). Covariates with the lowest p-values were

evaluated by testing linear or power relationships. To avoid the inclusion of highly correlated covariates on the same model parameter, visual inspection and as well as linear and rank biserial correlations were used. Each intermediate model was evaluated and compared with the others in terms of Akaike information criterion (AIC) and Bayesian Information Criterion (BIC) scores,⁴⁷ goodness of fit (GOF) plots, prediction corrected visual predictive check (pcVPC),⁴⁸ significance level of the included covariates with Wald test⁴⁹ (significance level was fixed to 1×10^{-3}), precision and correlations of estimated parameters,⁴⁷ diagnostic plots⁴⁷ and Shapiro–Wilk test (significance level was fixed to 1×10^{-3}) to check the normality assumption on model residual unexplained variability (RUV).⁵⁰

2.4 | Evaluation of model applicability to inform personalized carboplatin dosing

Predictive performance of the final carboplatin PK-PD model of myelosuppression was evaluated to assess its applicability in supporting individualized dosing in real-world scenarios. The objective was to determine whether the model could predict PLT and NT responses to carboplatin treatment at the individual patient level, and, thereby, inform dosing decisions in a setting where patient observations become sequentially available across treatment cycles.

To this aim, a retrospective evaluation was performed on the test set of 210 patients. For each patient and each treatment cycle i , patient data collected until cycle $i - 1$ were leveraged to estimate patient-specific PK-PD parameters via a full-Bayesian inference approach

using the developed pop-PK-PD model as prior information. Differently from Bayesian Maximum a posteriori (MAP) estimation,^{51,52} this approach does not estimate only the most likely set of patient parameters but generates the entire posterior distribution of the patient parameters (Panel A, Figure 2). Using this distribution, simulations were performed to predict PLT and NT profiles for cycle i , incorporating both uncertainty in the patient-specific parameter estimates based on data available until the $i - 1$ cycle, and RUV from the observation

model.^{53,54} Thus, for each time point of cycle i the model generated a probabilistic forecast for PLT and NT outcome. The goodness of model predictions was assessed by verifying whether observed PLT and NT values at the $i - th$ cycle fell within model predictive interval/ prediction probability distribution (Panel B, Figure 2). A detailed description of the simulation methodology and the statistical structure used to compute the 90% prediction intervals (90%PI) is provided in the Supplementary Material S.1.

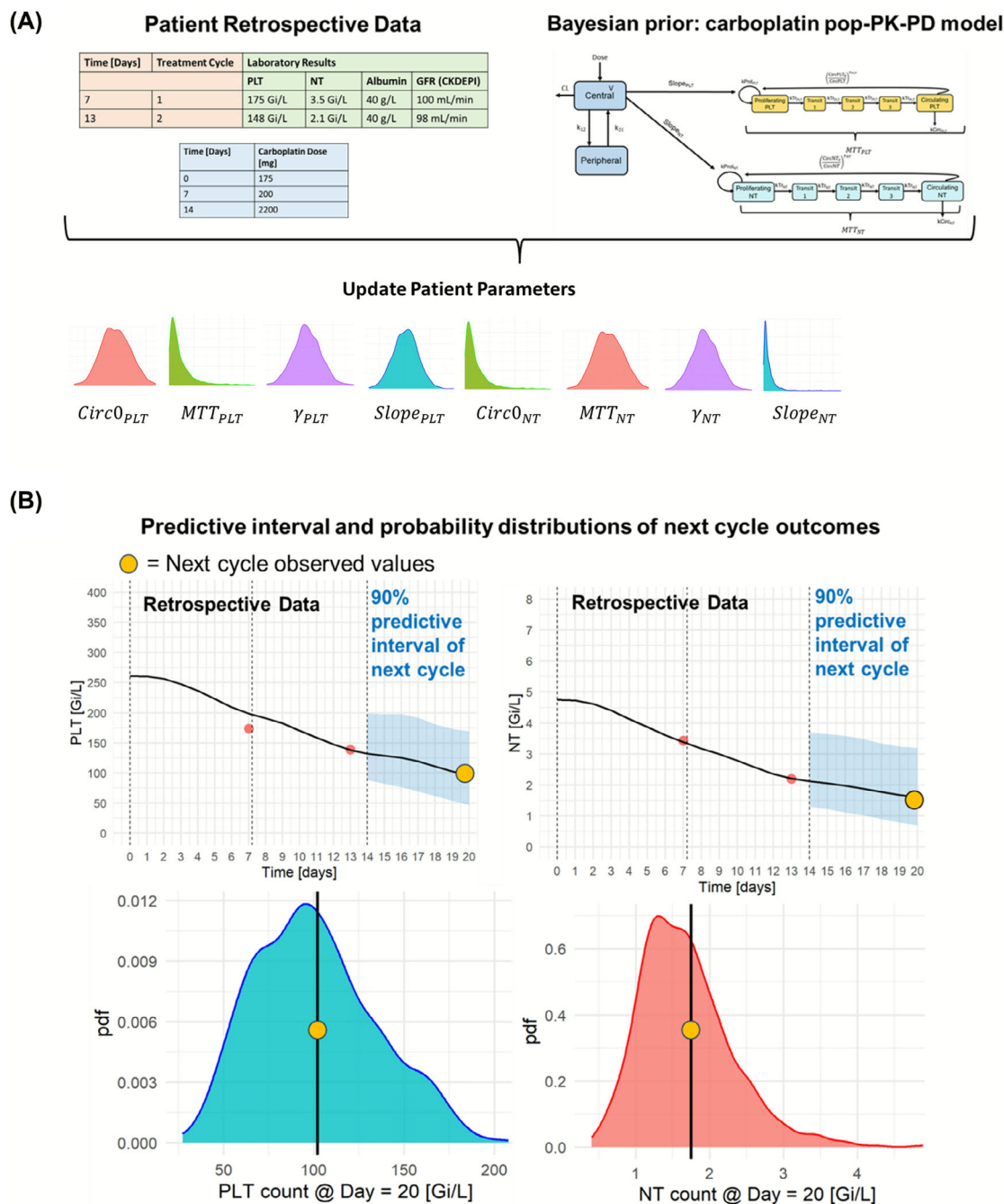


FIGURE 2 Representation of the evaluation framework for assessing the predictive performances of the developed pop PK-PD model. Panel A: patient data collected until cycle $i - 1$ are leveraged to estimate patient parameters with a full-Bayesian approach using the developed pop-PK-PD model as prior information. Panel B: estimation uncertainties and RUV are propagated to predict the outcomes of each time point in the $i - th$ cycle. To assess the goodness of model predictions, it was checked if the PLT and NT values collected at the $i - th$ cycle fall within the model predictive interval and at which percentile of the prediction probability distribution they belong.

2.5 | Software

Raw data manipulation and exploration were performed with R 4.2.1 and R-Studio. Monolix 2023R1 (Lixoft SAS, a Simulations Plus company) was used for the model development and the Bayesian estimation of individual parameters. Simulx 2023R1 (Lixoft SAS, a Simulations Plus company) and R were integrated through the Lixoft Connectors R package to perform model simulations using Bayesian posterior distribution of the model parameters for each patient at each treatment cycle.

3 | RESULTS

3.1 | Final pop-PK-PD model of carboplatin induced-myelosuppression

Different covariate models leveraging the covariates in Table 1 were tested on PLT and NT parameters to partially explain IIV (details on intermediate steps in Supplementary Materials S2). The final covariate model (Equation S2.1 of Supplementary Materials S2) is described in Figure 3, while Table 2 reports the parameter estimates as well as

their estimation error (RSE%). Serum albumin and eGFR estimated with CKD-EPI were included with a power model. A linear model was adopted for describing the progressive increase of carboplatin myelosuppression of PLT with the increase of treatment cycles. Log-linear effect was adopted for categorical covariates (i.e., paclitaxel comedication, pemetrexed comedication, oesophageal/gastric cancer diagnosis). All covariate coefficients were adequately estimated (RSE% < 30%, p-value of Wald test < 0.01, Table S2.1) with low estimation correlations (Figure S2.1).

Figure 4 illustrates the main graphical diagnostics for the final model. Panel A and B contain pcVPC for PLT and NT, respectively. Differently, in panels C and D, model predictions vs. observations are reported. Further model diagnostics on residuals and IIV are reported in Supplementary Materials S2.

3.2 | Physiological interpretation of final model covariates

Once the final model was identified, the subsequent analysis was conducted to assess the impact of the selected covariates on PLT and NT. To this end, the simulations reported in Figure 5 were done for a

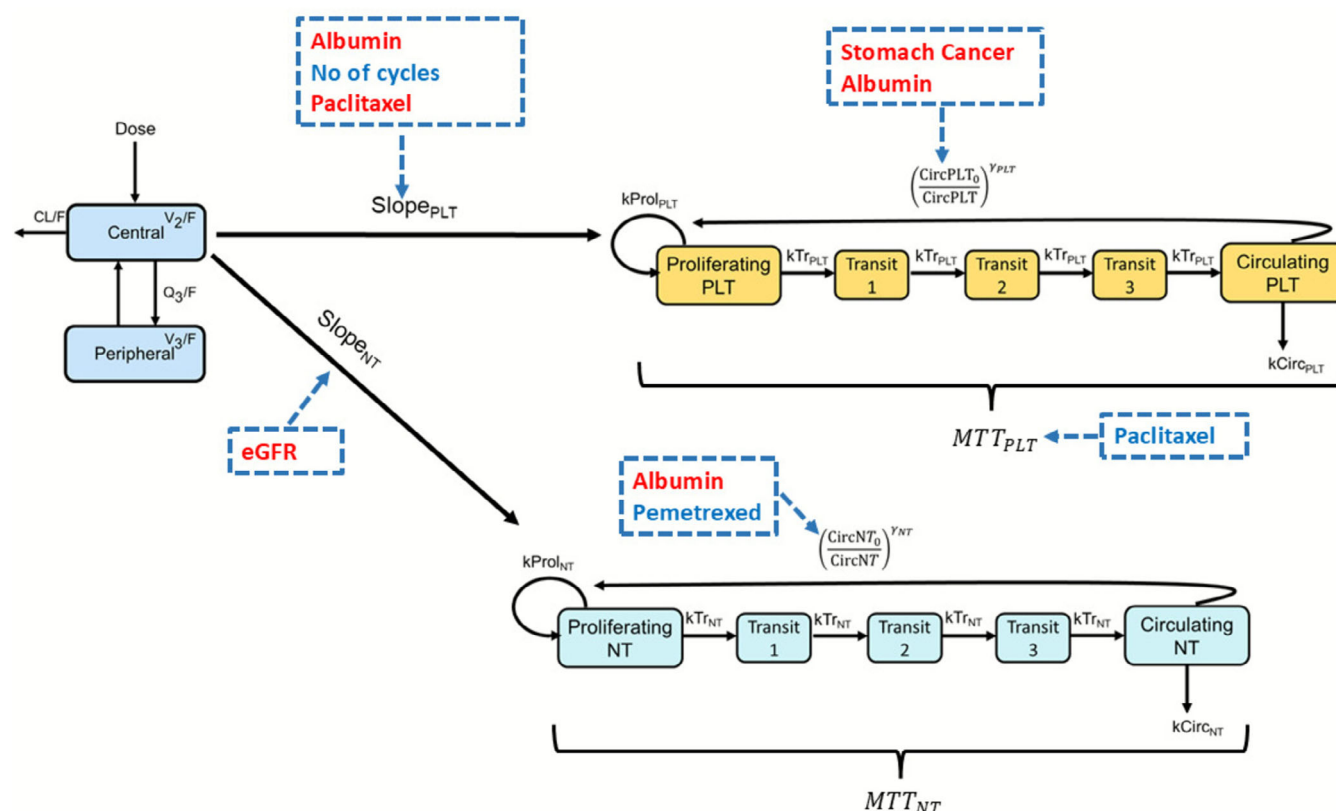


FIGURE 3 Final covariates model. Red: covariates whose effect was estimated with a negative coefficient, blue: covariates whose effect was estimated with a positive coefficient. As detailed in supplementary materials S2, serum albumin and GFR estimated with CKD-EPI were included with a power model. A linear model was adopted for describing the progressive increase of carboplatin myelosuppression of PLT with the increase of treatment cycles. Log-linear effect was adopted for categorical covariates (i.e., paclitaxel comedication, pemetrexed comedication, oesophageal/gastric cancer diagnosis).

TABLE 2 Carboplatin pop-PK-PD model parameters. Estimated values and relative standard error (RSE %) are reported. Residual unexplained variabilities (RUV) and IIV are reported in terms of standard deviations (ω and σ , respectively).

Parameter [units]	Description	Value [RSE %*]
Structural parameters		
$CircO_{PLT}$ [Gi/L]	Baseline PLT count	293 [2.0%]
$\beta_{gastric/esophageal}$ on $CircO_{PLT}$ [-]	Shift on $CircO_{PLT}$ in patient with oesophageal/gastric cancer diagnosis	-0.16 [17.0%]
$\beta_{Albumin}$ on $CircO_{PLT}$ [-]	Effect of the shift of serum albumin concentration with respect to a reference value of 44 g/L* on $CircO_{PLT}$	-0.503 [11.6%]
MTT_{PLT} [days]	Mean maturation time of PLT	6.27 [1.8%]
β_{PTX} on MTT_{PLT} [-]	Effect of paclitaxel co-medication on MTT_{PLT}	0.306 [10.2%]
γ_{PLT} [-]	Magnitude of PLT feedback mechanism	0.188 [3.4%]
$Slope_{PLT}$ [L/mg]	Inhibition of PLT proliferation due to carboplatin concentration	0.282 [6.8%]
$\beta_{Treatment}$ on $Slope_{PLT}$ [-]	Effect of treatment cycle on $Slope_{PLT}$	0.756 [8.54%]
$\beta_{Albumin}$ on $Slope_{PLT}$ [-]	Effect of the shift of serum albumin concentration with respect to a reference value of 44 g/L* on $Slope_{PLT}$	-0.26 [29.2%]
β_{PTX} on $Slope_{PLT}$ [-]	Effect of paclitaxel comedication on $Slope_{PLT}$	-0.562 [11.1%]
$CircO_{NT}$ [Gi/L]	Baseline NT count	5.16 [2.56%]
$\beta_{Albumin}$ on $CircO_{NT}$ [-]	Effect of the shift of serum albumin concentration with respect to a reference value of 44 g/L** on $CircO_{NT}$	-0.471 [18.3%]
$\beta_{Pemetrexed}$ on $CircO_{NT}$ [-]	Effect of pemetrexed comedication on $CircO_{NT}$	0.355 [10.9%]
MTT_{NT} [days]	Mean maturation time of NT	6.24 [3.2%]
γ_{NT} [-]	Magnitude of NT feedback mechanism	0.074 [5.51%]
$Slope_{NT}$ [L/mg]	Inhibition of NT proliferation due to carboplatin concentration	0.509 [4.45%]
$\beta_{GFR\ CKDEPI}$ on $Slope_{NT}$ [-]	Effect of the shift of estimated GFR with CKDEPI equation with respect to a reference value of 100 mL/min*** on $Slope_{NT}$	-0.602 [11.9%]
IIV random effects (ω)		
$\omega_{CircO_{PLT}}$	-	0.268 [4.1%]
$\omega_{MTT_{PLT}}$	-	0.190 [7.2%]
$\omega_{\gamma_{PLT}}$	-	0.358 [7.67%]
$\omega_{Slope_{PLT}}$	-	0.547 [4.71%]
$\omega_{CircO_{NT}}$	-	0.326 [5.08%]
$\omega_{MTT_{NT}}$	-	0.322 [8.53%]
$\omega_{\gamma_{NT}}$	-	0.626 [6.97%]
$\omega_{Slope_{NT}}$	-	0.624 [5.40%]
Correlation between IIV random effects (ρ)		
$\rho(CircO_{PLT}, CircO_{NT})$	-	0.440 [13%]
$\rho(Slope_{NT}, \gamma_{NT})$	-	0.725 [5.76]
Residual Unexplained Variability		
$\sigma_{PLT, Additive}$	-	5.92 [16%]
$\sigma_{PLT, Proportional}$	-	0.218 [2.96%]
$\sigma_{NT, Proportional}$	-	0.321 [1.89%]

Notes:

*RSE% of parameter estimates were derived from the Fisher Information Matrix, which was estimated using the stochastic approximation method implemented in Monolix.

**The reference value of 44 g/L for serum albumin was arbitrarily selected in the middle of normal physiological range (typically 35–50 g/L).

***The reference value of 100 mL/min for CKDEPI-eGFR was arbitrarily selected within the normal physiological range (typically 90–120 mL/min).^{55–57}

typical gastric/oesophageal cancer patient treated with a combination of carboplatin, paclitaxel and radiotherapy and varying each covariate within the observed range in the population used for model

development (i.e., the 95%CI reported in Table 1). Simulations were performed by using the standard carboplatin dosage according to the Calvert equation (i.e., $Dose = AUC_{Target} * (25 + GFR)$ with $AUC_{Target} = 2$

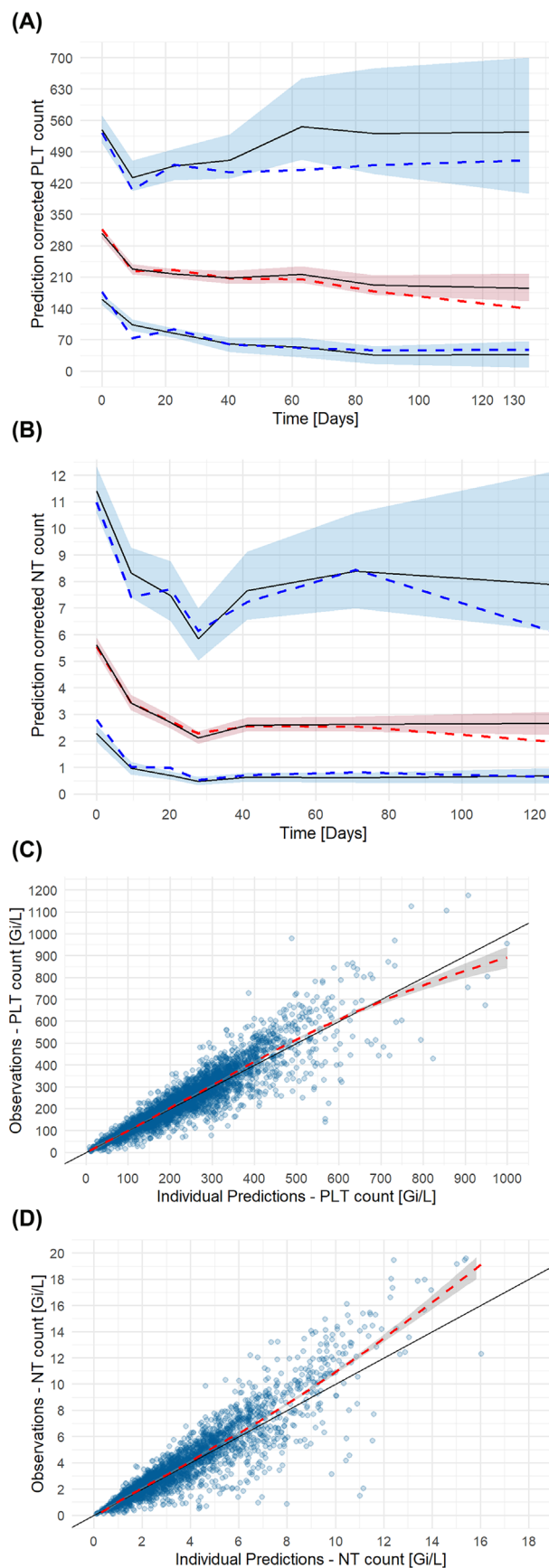


FIGURE 4 Diagnostic plots of the final pop-PK-PD model for carboplatin-induced myelosuppression. Panels A and B illustrate prediction corrected VPC (pcVPC) for PLT and NT, respectively. Solid black lines represent 2.5th, 50th and 97.5th percentiles of training data. Dashed lines are 2.5th, 97.5th (blue) and 50th (red) percentiles of simulations obtained with the final model. Shaded areas are 95% CI of the simulated upper/lower percentiles (blue) and the median (red). Panels C and D reports the scatterplots of model prediction vs. observations for PLT and NT, respectively. Black line is the identity line, the red dashed line and grey shaded are the spline and its CI representing the trend of the points.

QW).^{8,15} In particular, the shift of serum albumin concentration with respect to a reference value of 44 g/L resulted in a significant effect on baseline PLT and NT count ($CircO_{PLT}$, $CircO_{NT}$) and on $Slope_{PLT}$. Albumin effect was estimated with a negative coefficient on both initial values of PLT and NT (Table 2). Therefore, if albumin is < 44 g/L, the baseline circulating levels (i.e., at day 0) of PLT and NT are higher (Panels A-B of Figure 5). Furthermore, albumin has a proportionally inverse effect on $Slope_{PLT}$, meaning that if albumin is < 44 g/L, PLT count lowers more rapidly (Panel A of Figure 5). The shift from a reference CKD-EPI eGFR of 100 mL/min had a significant impact on $Slope_{NT}$ with a negative β as well. A lower renal function associated to a more aggressive reduction of NT count (Panel C of Figure 5). Interestingly, it was found that carboplatin myelosuppressive effect on PLT ($\beta_{Treatment\ on\ Slope_{PLT}} = 0.756$) increased in time (further details in Supplementary Materials S2). As concerns the effect of concomitant medications, only paclitaxel (PTX) and pemetrexed (PEM) chemotherapies had a significant impact on model covariates. In particular, PTX lowered the carboplatin effect on $Slope_{PLT}$ and increased PLT maturation time (Panel D of Figure 5). Conversely, patients following concomitant PEM administration had a higher baseline NT level.

3.3 | Evaluation of model applicability to inform personalized carboplatin dosing

Model predictive performances were evaluated on the test dataset of 210 patients (Figure 2) to assess the model capability to exploit patient's retrospective data to infer future outcomes (i.e., next-cycle PLT and NT values) in the same patient. As illustrated in Figure 6, almost all the PLT and NT observations fell within model 90%PI at each treatment cycle. In particular, the percentage of observed PLT and NT outside this range was always maintained below 15% at each treatment cycle. At baseline, 89% of PLT observations and 85% of NT observations fell within the model 90%PI. As intuitively expected, as more data were collected from a given patient (e.g., over successive treatment cycles), the accuracy of model predictions improved, with accuracy >90% from cycle 2 onwards. Furthermore, the posterior predictive check analysis (Figure S2.6 of Supplementary Materials S2) highlighted that model predictions on the test set are unbiased for all treatment cycles.

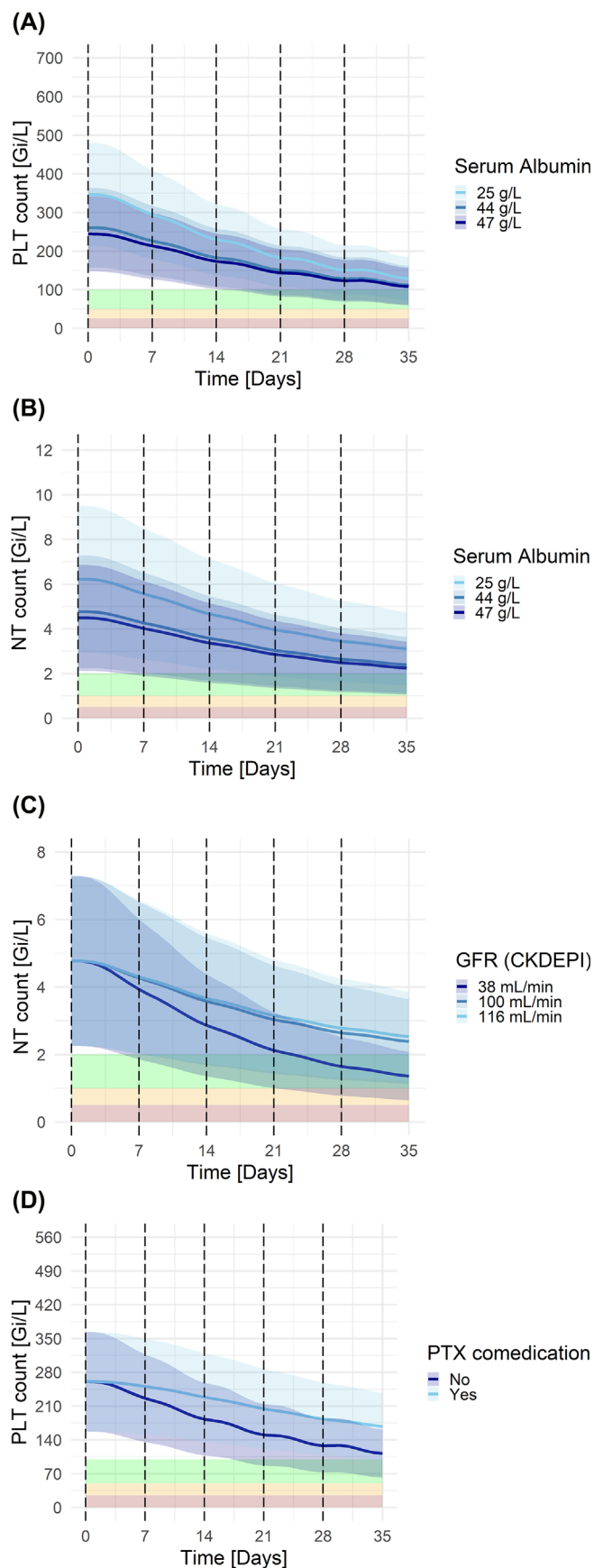


FIGURE 5 Simulations on a typical gastric/oesophageal cancer patient to explain the effect of the included covariates. Panels A and B show the effect of different serum albumin levels on PLT and NT count, respectively. Panels C and D focus on the effect of GFR (estimated with CKD-EPI equation) and paclitaxel co-medication on NT and PLT, respectively. Shaded areas represent the 90%CI of the predictions. Dashed lines indicate carboplatin (and paclitaxel in panel D) administrations. Green areas indicate $PLT \in (50, 100]$ Gi/L or $NT \in (1, 2]$ Gi/L (i.e., grade 1 and 2 thrombocytopenia and neutropenia, respectively). Yellow areas stand for $PLT \in (25, 50]$ Gi/L or $NT \in (0.5, 1]$ Gi/L (i.e., grade 3 thrombocytopenia and neutropenia). Finally, red areas indicate $PLT \leq 25$ Gi/L or $NT \leq 0.5$ Gi/L, which represent severe thrombocytopenia and neutropenia, respectively.

3.4 | Example of model-based dose suggestions

In the previous section, the capability of the proposed PK-PD model to predict individual PLT and NT responses to carboplatin based on data collected in prior treatment cycles was demonstrated. In this section, the applicability of the model to support MIPD of carboplatin is exemplified through a retrospective evaluation conducted through simulation on the lung cancer patient cohort ($n = 117$) of the test population. The goal of the proposed MIPD approach is to identify, for each patient and treatment cycle, the highest tolerable carboplatin dose that minimizes the risk of severe haematological toxicity (i.e., grade ≥ 3 thrombocytopenia and/or neutropenia). To this end, for each patient, the following procedure was implemented:

- For each cycle i , the individual posterior distribution of PK-PD model parameters was estimated via full Bayesian inference using only the data up to cycle $i-1$ (as detailed in the “Evaluation of model applicability to inform personalized carboplatin dosing” section).
- Based on these patient-specific parameter distributions, the model was used to identify an ‘optimal dose’ for cycle i , while keeping the doses for previous cycles fixed to the actual clinical values.
- The ‘optimal dose’ was defined as the highest carboplatin dose satisfying the following safety criteria:
 - i. $\text{prob}(\text{observing any grade 4 toxicity at nadir}) < 0.1$
 - ii. $\text{prob}(\text{observing any grade 3 or 4 at monitoring day}) < 0.1$
 - iii. PLT and NT recovery from nadir.

A grid search strategy was implemented to explore possible dose adjustments within a maximum of $\pm 50\%$ of the standard carboplatin dose (i.e., $\text{Dose} = \text{AUC}_{\text{target}} * (\text{GFR} + 25)$, with $\text{AUC}_{\text{target}} = 5$) in 10% increments (i.e., the following candidate doses were tested: 50%, 60%, 70%, 80%, 90%, 100%, 110%, 120%, 130%, 140% and 150% of the Calvert-based dose).

The model-based dosing strategy was compared through simulation with routine clinical practice, which relies on the AUC5 criteria from the Calvert equation but may also involve adjusting the target AUC if patients develop toxicity not reflected by changes in eGFR. For this comparison, the final posterior distributions of the individual

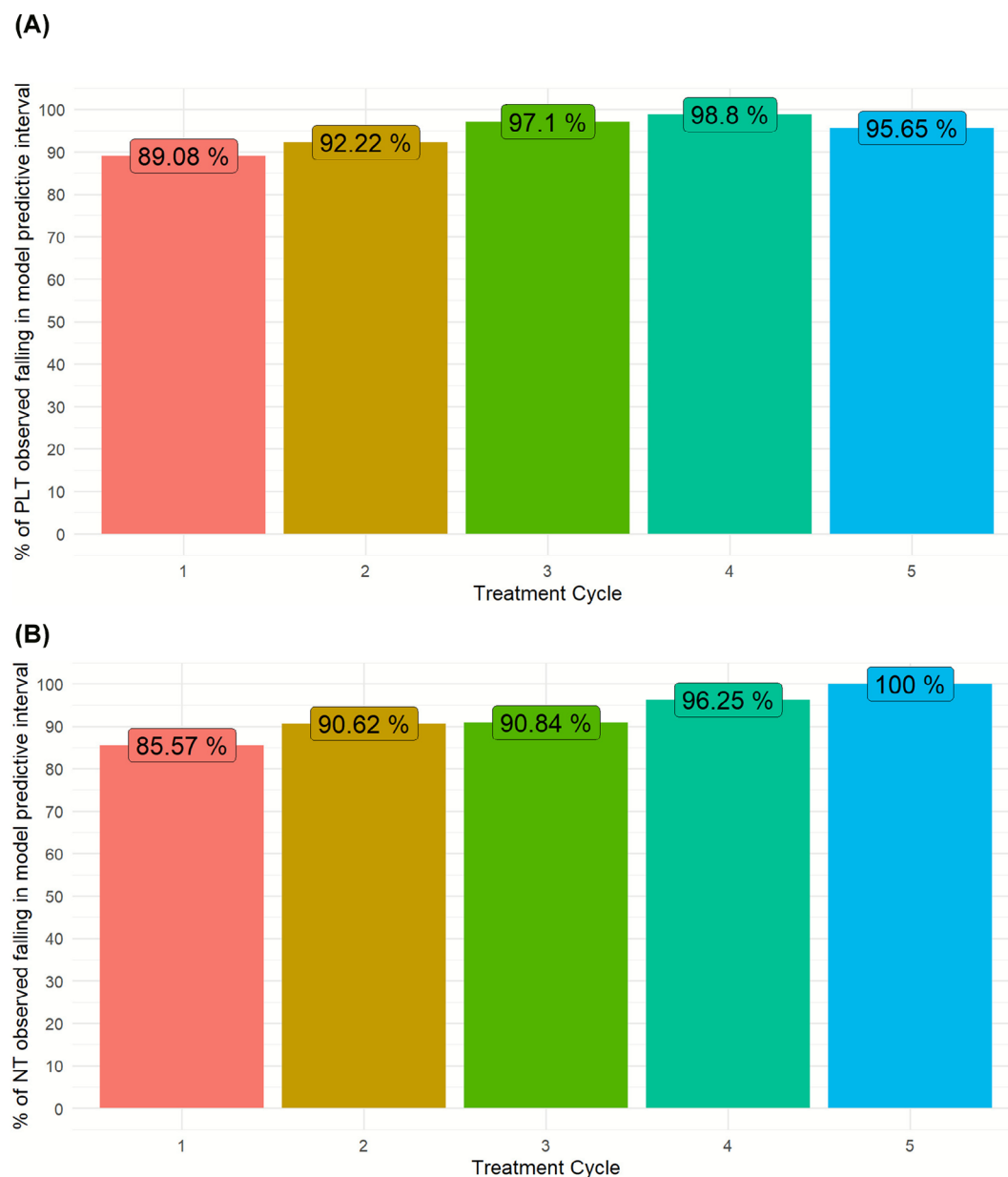


FIGURE 6 Assessment of model predictive performances. For each treatment cycle, the percentage of PLT observations (panel A) and NT observations (panel B) falling within the model 90%PI are reported. These percentages indicate how well the model predictions align with the observed data.

PK-PD parameters (inferred using all the individual data for each patient) were used to compute, for each cycle:

- the probability of $\geq$ grade 3 haematological toxicity at the end of treatment cycle, i.e., day 21 after dose administration;
- the probability of $\geq$ grade 4 toxicity at nadir;

under both the clinically administered dose and the optimal (MIPD) dose.

Key results from this analysis are summarized in Tables S3.1-S3.4 in Supplementary Material S3 and exemplified in Figure S3.1. Notably:

- In cycle 1, the model suggested a higher dose than the AUC5-based dose in 83% of patients, while in later cycles it recommended a dose reduction in over 50% of patients (Table S3.1).
- Following both treatment strategies, generally, few patients had a high risk (probability ≥ 0.1) of severe toxicity, both at nadir and at the monitoring day, across all cycles (Table S3.2).
- Starting from cycle 2, the proportion of patients at high risk of severe toxicity was consistently lower under the MIPD protocol compared to the clinical dosing (Table S3.2). In particular, the number of patients who would have experienced high-probability toxicity under AUC5 clinical dosing but not under the MIPD dose (i.e., toxicity preventable with MIPD) was substantial and clinically

relevant from cycle 2 onwards (Table S3.3). Conversely, the number of patients who would have developed toxicity only under MIPD but not under standard AUC5 dosing was very low and decreased across cycles, reaching zero at cycle 4 (Table S3.4).

These findings indicated that the model was able to identify dose levels that maintain treatment within a clinically safe toxicity range, achieving the highest tolerable dose for each patient. This demonstrates the feasibility of the approach and its clinical relevance, as it decreases the risk of severe toxicity compared with fixed or non-individualized dosing strategies. The only exception was during the first treatment cycle, where predictions based solely on baseline blood counts were less informative and were associated with a higher risk of toxicity.

4 | DISCUSSIONS

In this retrospective single-centre study, a pop-PK-PD model was developed and validated to support carboplatin dose personalization for the most common tumour types for which this compound is a standard treatment (i.e., lung, gynaecological and oesophageal/gastric neoplasm). Such methodology could improve clinical outcomes compared to the current carboplatin dosing strategy, based on the Calvert equation. Calvert dosing computes the carboplatin dose for achieving a target exposure level through patient CG-GFR estimates.^{7-9,15} It has been reported that Calvert dosing cannot achieve adequate safety and efficacy results in all patients, resulting in severe hematologic toxicities^{5,6} (i.e., grade 3/4 thrombocytopenia and/or neutropenia according to CTCAE definition)¹⁶ and tumour progressions.⁵⁸⁻⁶⁰ Consequently, the need for more personalized dosing approaches to maximize carboplatin treatment efficacy (i.e., avoiding underdosing) and reduce its toxicity was underlined by several works,^{11-14,58,61,62} and this work aimed to address this task.

The basic structure of the developed carboplatin-induced myelosuppression pop-PK-PD model leveraged both the carboplatin pop-PK model implemented by Ekhardt et al.²⁶ and the semi-mechanistic PD model of chemotherapy-induced myelosuppression developed by Friberg et al.²⁷ The final pop-PK-PD model developed on a training set cohort of 580 patients is schematized in Figure 3. Table 2 reports the obtained model parameter estimates, which were obtained with low estimation errors and are aligned with previous literature works on the modelling of chemotherapy-induced myelosuppression, thus confirming the robustness of the obtained results.^{33,63-68}

Serum albumin was the most predictive covariate of carboplatin myelotoxicity. In patients with relatively low albumin values (~ 25 g/L, i.e., the lower bound of observed range), the model predicted higher baseline circulating levels (i.e., at day 0, $CircO_{PLT}$ and $CircO_{NT}$ model parameters) of PLT and NT (Figure 5, Panels A-B). This finding seems coherent with the physiology of inflammation and cancer, which are characterized by a reduction of albumin serum levels and higher PLT and NT counts.^{69,70} Analogously, when serum albumin is < 44 g/L, the

model predicted a stronger cytotoxic effect of carboplatin on PLT ($Slope_{PLT}$). This result is coherent with albuminuria episodes related to a reduction of renal function, which can occur during chemotherapy.^{28,71} Interestingly, serum albumin resulted in a more significant effect (i.e., lower p-value of Pearson Correlation test) than CKD-EPI eGFR and serum creatinine on $Slope_{PLT}$. Conversely, a greater decrease in CKD-EPI eGFR from the reference value of 100 mL/min (chosen as a physiological benchmark) is associated with a stronger impact of carboplatin concentration on the proliferation rate of normal tissues (i.e., higher values of the $Slope_{NT}$ model parameter). This finding is consistent with literature reports linking greater carboplatin exposure to reduced renal function and increased myelotoxicity.^{28,72,73} The identification of eGFR as a significant predictor of carboplatin's impact on myelosuppression also highlights a limitation in accurately determining patient-specific PK parameters. This aspect is mainly due to the lack of measurements for carboplatin concentration levels in blood during treatment. In our study, only patient covariates (i.e., age, weight, serum creatinine and gender) were leveraged to estimate PK parameters according to the Ekhardt model.²⁶ The Ekhardt model was developed in a Dutch oncology population, which is generally demographically comparable to the populations of this study. However, cohorts included in this work comprise patients with slightly more advanced age (95th percentile: 84 vs. 75 years in the Ekhardt population) and higher serum creatinine values (95th percentile: 139 vs. 124 μ mol/L). Moreover, the Ekhardt population only included patients with eGFR > 55 mL/min. Such demographic differences, particularly the older age and reduced renal function of our patients, may have influenced the predictive performance and transferability of the model-predicted concentration-time profiles to our population, as recently reported by Kicken et al., who showed that inaccurate estimation of renal function can lead to deviations between predicted and achieved carboplatin exposure.⁷⁴ However, the absence of direct carboplatin concentration measurements in our dataset represents a study limitation, as it constrains the ability to disentangle PK from PD variability and restricts confidence in extrapolating model predictions, particularly in patients with impaired renal function.

Among the concomitant immune- and chemotherapeutic treatments, only paclitaxel and pemetrexed were predictive of carboplatin myelotoxicity (Table 2 and Panel D of Figure 5). Interestingly, although both capecitabine and etoposide have a myelosuppressive effect,³¹⁻³⁴ they resulted less significant than paclitaxel and pemetrexed (i.e., higher p-value of ANOVA test). Such a finding may be related to the small percentage of patients in the training set (Table 1) receiving these co-medications. Another covariate that was expected to be impactful but was ultimately excluded from the final model is the serum level of LDH. Although it is occasionally used as a surrogate biomarker of tumour burden or disease aggressiveness in specific cancers, such as small-cell lung cancer and germ cell tumours,^{39,41} its prognostic value is highly context-dependent. Moreover, LDH is not a direct indicator of bone marrow reserve or of the patient's susceptibility to chemotherapy-induced myelosuppression, which was the primary endpoint of our study. These factors may explain the lack of predictive value of LDH for haematological toxicity in our analysis.

Interestingly, the final model predicts a linear increase in carboplatin myelosuppressive effect on PLT after multiple treatment cycles, but not on NT. NT recovery is influenced by multiple factors (e.g., comorbidities, baseline immune status, supportive interventions, infections), which may obscure such longitudinal patterns. The observed longitudinal effects on PLT are coherent with previous literature studies that found a reduction of bone marrow activity after multiple chemotherapy administrations due to a cumulative toxicity effect.^{75–77} While a saturable Emax-type function was also tested, it did not improve model performance ($\Delta\text{BIC} = +38$, $\Delta\text{AIC} = +21$). Considering the limited number of cycles per patient and the characteristics of the available data, the linear formulation offered a more appropriate balance between fit quality and model parsimony.

To evaluate the applicability of the final carboplatin PK-PD model in a MIPD setting where patient observations become sequentially available, the final model was tested on a test dataset of 210 patients of the same centre. The model showed the lowest predictive performances in the first treatment cycle, in which patient outcomes are inferred only by leveraging baseline characteristics. As more data are collected for a given patient (e.g., over successive treatment cycles), the accuracy of model predictions improved. While predictions at baseline may be improved by future dedicated covariate analyses, this finding, along with the analysis reported in Supplementary Materials S2, confirmed that the developed model is able to make accurate and unbiased predictions of next-cycle patient outcomes. The use of the model to develop MIPD strategies was exemplified through a retrospective simulation study conducted on the lung cancer test cohort ($n = 117$). For each patient and treatment cycle, the model was employed to identify an individualized “optimal” dose, defined as the highest carboplatin dose satisfying predefined safety criteria. These model-based dosing recommendations were subsequently compared with the doses actually administered in clinical practice, which were based on the Calvert formula (AUC5) and potentially adjusted by clinicians according to patient response. The results indicated that the model-based approach could effectively reduce the number of patients at high risk of severe haematological toxicity from cycle 2 onward, while enabling both dose escalation and reduction in a personalized manner. An example that such toxicity-based dosing is feasible and can effectively be implemented in the clinic was shown in Dose-Adjusted EPOCH trial. Here, the baseline chemotherapy dose is based on the severity of disease whereas doses of etoposide, doxorubicin and cyclophosphamide are adjusted during subsequent cycles based on the neutrophil nadir, monitored twice weekly.⁷⁸ A similar monitoring strategy can be implemented for future prospective evaluation of the carboplatin MIPD approach.

These findings provide a practical illustration of how model-based dose individualization could be translated into clinical workflows. However, despite such promising applications, MIPD is still uncommonly applied in oncology. Most existing chemotherapeutic MIPD studies have primarily relied on PK exposure metrics (cycle-based AUC or $C_{ss,avg}$) to link drug concentrations to efficacy or toxicity outcomes. A recent review provides an overview of these strategies mainly relating to busulfan, cyclophosphamide, anthracyclines and

topoisomerase inhibitors.⁷⁹ In contrast, the proposed MIPD strategy uses haematological counts (neutrophils, platelets) as biomarkers for individualized dosing. Such toxicity- or biomarker-guided MIPD approaches have been explored,^{80–82} but remain less established in clinical practice.

Although the model showed good predictive performances and feasibility for MIPD implementation, several limitations remain and need to be addressed to support clinical translation. First, the evaluation of the model applicability was performed on a test set of 210 cancer patients of Amsterdam UMC. To confirm its generalizability and robustness, it could be useful to extend its evaluation on a population from another clinical centre. Second, in this real-world dataset, it was not feasible to fully differentiate among each cancer diagnosis, prior therapies and staging line as covariates in the PK-PD analysis. Future studies should focus on specific diseases and should include structured information on prior (chemo)therapy and disease grades to refine the predictive value of the PK-PD relationships and improve individualized treatment strategies. Third, the lack of PK sampling in the current clinical workflow limits the precision of model prediction. As discussed before, this aspect prevents a robust estimation of patient-specific PK parameters and was partially compensated by including some biomarkers of renal function (i.e., CKD-EPI eGFR) as covariates on the carboplatin myelosuppressive effect. However, if carboplatin concentration values were to be monitored prospectively, such additional measurement could further improve model prediction by reducing RUV and random IIV, in line with previous studies.^{33,63–68} Also, the modelling framework should be expanded to incorporate a mechanistic description of how concomitant chemotherapeutics affect myelosuppression, rather than treating them as binary covariates,^{33,65} which would allow assessment of the effect of multiple myelosuppressive drugs simultaneously. Additionally, in this study, serum creatinine was used to estimate GFR for carboplatin dosing. However, creatinine levels are influenced by non-renal factors such as age, sex and muscle mass, which may lead to inaccurate renal function estimates in oncology patients, particularly those with cachexia or altered body composition. Cystatin C, a low-molecular-weight protein produced by all nucleated cells, is less affected by these variables and may offer a more accurate assessment of renal function. Studies have shown that cystatin C, alone or in combination with creatinine, improves GFR estimation and may better predict outcomes in oncology and critically ill populations.^{83,84} Although its routine clinical use is still limited by cost and availability, further research could support its implementation in carboplatin dosing for cancer patients.

Another major limitation concerns the uncertainty of model predictions. Specifically, the Bayesian prediction intervals for PLT and NT counts may span a range wide enough to cross multiple CTCAE toxicity grades, thereby limiting the ability to derive precise and actionable clinical decisions from model outputs. Indeed, although the uncertainty on individual model parameters tends to decrease as more patient-specific observations are collected across treatment cycles, the overall prediction intervals remain wide due to the dominant contribution of RUV. This limits the accuracy of forward predictions and may reduce the benefits of longitudinal data assimilation in clinical

decision-making. These considerations highlight the importance of future efforts aimed at reducing RUV, for instance, through optimized sampling strategies or the integration of PK measurements, to improve the predictive performance and clinical utility of population models for MIPD applications.

It is also important to note that the implemented MIPD strategy was limited to a toxicity-focused approach based only on thrombocytopenia and neutropenia, where the goal was to administer the highest tolerable dose without inducing severe haematological toxicity events (i.e., CTCAE ≥ 3). Neutropenia can be a predictive and prognostic marker for survival for many classical cytotoxic agents in lung, gynaecological and oesophageal/gastric neoplasms.^{85–87} In hematologic disease, the dose-adjusted EPOCH study showed that patients benefitted from dose adjustment to achieve a nadir absolute neutrophil count below $0.5 \times 10^9/L$.⁸⁸ Although a toxicity-based dosing strategy has not been studied for carboplatin-based chemotherapy regimens in solid tumours yet, a similar benefit can be expected if dose modifications are used to reach the maximum tolerated dose, accepting mild-to-moderate myelosuppression (CTCAE grade ≤ 2). Of course, a crucial next step toward clinical implementation is the integration of treatment efficacy metrics, either short-term (e.g., tumour markers, ctDNA or tumour shrinkage based on imaging data) and/or long-term (e.g. overall and event-free survival, OS and EFS, probability) within the modelling framework. While OS/EFS data were not yet mature at the time of model development, and longitudinal efficacy biomarkers could not be extracted from the EHR system, a key direction for future development of the framework will be the inclusion and modelling of one or more efficacy biomarkers. In particular, access to longitudinal efficacy data, such as tumour markers or imaging-based assessments, would enable the implementation of a MIPD strategy that directly balances therapeutic benefit and toxicity, thereby overcoming the current limitations of a toxicity-only approach. Finally, the proposed modelling framework focused only on haematological toxicity that is currently the most frequent and dose-limiting adverse event for carboplatin. Non-hematologic adverse events, such as nephrotoxicity, nausea/vomiting, peripheral neuropathy/ototoxicity and hypersensitivity reactions, occur less frequently and are generally not strongly correlated with either the dose or exposure nor with the depth or duration of myelosuppression.^{89,90} Nephrotoxicity with carboplatin is uncommon relative to cisplatin⁹¹ and is influenced mainly by baseline renal function, prior cisplatin exposure, hydration status and co-medications. Importantly, renal function is already incorporated into carboplatin dosing through the CG-GFR estimates,^{7,8,10,15} and in our current model, eGFR is explicitly used as a covariate to predict the exposure–thrombocytopenia relationship. Nausea/vomiting with carboplatin is typically moderate-emetogenic and largely mitigated by guideline-concordant prophylaxis, with no consistent relationship to hematologic nadirs.⁹¹ Hypersensitivity reactions are more closely related to cumulative exposure (e.g., after multiple cycles) and are not predicted by myelosuppression; if biomarker-guided dosing enables longer treatment duration at tolerable counts, vigilance for late hypersensitivity remains warranted.

Given these considerations, our current MIPD framework does not account for such toxicities, but they should still be monitored independently in clinical settings. Looking ahead, the development of multi-endpoint MIPD strategies that integrate both hematologic and non-hematologic outcomes may offer a more comprehensive approach to treatment individualization and toxicity management. In such complex scenarios, with multiple efficacy and toxicity endpoints to simultaneously optimize, the integration of Artificial Intelligence algorithms such as reinforcement learning (RL)^{92–95} could be particularly valuable, enabling real-time optimization of treatment trajectories across cycles. Current efforts are moving in this direction. These advancements will be instrumental in enabling the clinical implementation of model-based dose individualization.

In conclusion, this work lays the foundation for a model-based approach to personalize chemotherapy and defines the roadmap of future developments and improvements for a clinical application of this patient-centric paradigm.

AUTHOR CONTRIBUTIONS

Alessandro De Carlo, Mirjam Crul, Tim Schutte, Paolo Magni and Imke Bartelink designed the study. Alessandro De Carlo, Harmen Huls and Imke Bartelink collected and curated the data. Alessandro De Carlo analyzed the data, and Elena Maria Tosca validated the analyses and updated the results. Alessandro De Carlo, Elena Maria Tosca, Medhat M. Said, Paolo Magni and Imke Bartelink contributed to the model interpretation, whereas Mirjam Crul, Tim Schutte, Lia van Zuijlen, Idris Bahce and Jan Buter contributed to the clinical interpretation of the results. Alessandro De Carlo and Elena Maria Tosca drafted the manuscript. All authors critically revised the manuscript and approved the final version.

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CONFLICT OF INTEREST STATEMENT

The authors had no conflicts of interest to disclose.

DATA AVAILABILITY STATEMENT

Data are available upon reasonable request from the corresponding author.

PATIENT CONSENT STATEMENT

NA.

PERMISSION TO REPRODUCE MATERIAL FROM OTHER SOURCES

NA.

CLINICAL TRIAL REGISTRATION

NA.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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